Olist 360: Sales, Retention, Delivery & Customer Intelligence Dashboard

Project Frame (Hiring-Ready)

Appendices

1. Executive summary.

Olist 360 is a business intelligence and customer analytics project designed to provide an executive-level view of marketplace performance across sales, customer retention, delivery operations, and customer experience. Built on a real-world e-commerce dataset covering nearly 100,000 orders from 2016 to 2018, the project consolidates multiple business dimensions—including orders, customers, products, sellers, payments, reviews, and geolocation—into a single decision-support framework.

The purpose of the project is to help stakeholders move from fragmented operational reporting to a more strategic understanding of what drives growth, where performance gaps exist, and which customer and product segments create the greatest long-term value. The analysis combines commercial performance tracking with customer intelligence and operational diagnostics, covering revenue and order trends, product and category performance, delivery efficiency, customer segmentation, retention analysis, repeat purchase behavior, and product combination patterns.

Key Business Highlights

- Growth is driven by a limited set of high-performing categories, regions, and customer groups, indicating clear concentration in revenue generation.
- Delivery performance plays a direct role in customer experience, with delayed orders consistently associated with weaker satisfaction signals.
- Customer value is unevenly distributed, creating a compelling case for retention-focused strategies aimed at high-value and repeat buyers.
- Basket-level purchasing patterns reveal practical cross-sell and bundling opportunities that can improve revenue per order.

Overall, the project delivers a business-facing analytics solution that helps leadership teams identify growth levers, strengthen customer retention, improve operational performance, and support more profitable marketplace decisions.

2.Background / Business Context

Olist is a Brazilian e-commerce marketplace that connects sellers to customers across multiple marketplaces. The business needs a 360-degree analytical view of sales performance, customer retention, delivery efficiency, and satisfaction in order to support strategic decisions and improve growth.
The dataset contains nearly 100k real-world marketplace orders from 2016 to 2018, including orders, payments, products, sellers, customers, and reviews.

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3. Problem Statement

The business lacks a unified analytical view that connects sales performance, customer loyalty, delivery operations, and satisfaction. Without this, it becomes difficult to identify the true drivers of revenue, understand retention behavior, and detect operational bottlenecks that negatively affect customer experience.

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4. Business Objectives & Questions

1. Analyze revenue and order trends over time

- How have revenue and order volume changed over time?
- Which periods and geographies drive revenue growth?
- Are there seasonal patterns in sales performance?

2. Evaluate product and category performance

- Which categories and products generate the highest revenue and order volume?
- Which categories underperform in sales, ratings, or operations?
- Which categories and regions require improvement?

3. Measure delivery performance and its impact on reviews

- What is the late delivery rate across regions, categories, and sellers?
- How do delivery delays affect customer satisfaction and review scores?
- Where are the main logistics bottlenecks?

4. Segment customers using RFM logic

- Which customer segments exist based on recency, frequency, and monetary value?
- Which segments contribute most to revenue?
- Which segments are high-value, and which are at risk?

5. Analyze retention using cohort methodology

- What are the retention rates of customer cohorts over time?
- Which cohorts retain customers best?
- How does retention evolve month after month?

6. Identify customer value concentration and repeat purchase behavior

- How concentrated is revenue among top customers?
- What share of customers are repeat buyers?
- How much revenue comes from repeat versus one-time customers?

7. Explore product combinations through basket analysis

- Which products or categories are commonly purchased together?
- What bundle and cross-sell opportunities can be identified?

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5. Stakeholders / Business Users

This project is designed for multiple business stakeholders across the e-commerce marketplace, each requiring a different analytical view to support decision-making. Because the Olist dataset captures marketplace activity across **orders, customers, products, sellers, payments, reviews, and delivery-related information**, the dashboard is relevant to both **commercial** and **operational** users.

A. Executive Leadership / Business Management

Executive stakeholders can use the dashboard to monitor overall marketplace performance, evaluate revenue drivers, understand customer retention patterns, and identify operational gaps that may affect growth and customer experience.

B. Sales and Category Managers

Sales and category teams can use the analysis to track revenue and order trends, compare category performance, identify top-performing products, and detect underperforming categories that may require pricing, assortment, or promotional adjustments.

C. Customer Retention / CRM / Marketing Teams

Customer-focused teams can use the dashboard to understand repeat purchase behavior, evaluate customer value concentration, monitor retention patterns, and identify high-value or at-risk segments through RFM and cohort-based analysis.

D. Operations and Logistics Teams

Operations users can use the dashboard to monitor delivery performance, track late delivery rates, identify logistics bottlenecks by category or geography, and assess how shipping outcomes influence customer satisfaction.

E. Seller Performance / Marketplace Operations Teams

Because the marketplace model includes multiple sellers and order fulfillment dimensions, seller management teams can use the dashboard to evaluate seller-related operational performance, delivery reliability, and quality signals such as review outcomes.

F. Customer Experience / Service Quality Teams

Teams responsible for customer experience can use the project to analyze review-score patterns, understand the relationship between service quality and satisfaction, and detect areas where delivery or order issues may be reducing trust in the marketplace.

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6. Data Source

Olist Brazilian E-Commerce Public Dataset by Olist (Kaggle):
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

The analysis is based on the **Brazilian E-Commerce Public Dataset by Olist**, a real-world anonymized marketplace dataset covering nearly **100,000 orders between 2016 and 2018**. It includes integrated business dimensions such as **orders, customers, products, sellers, payments, reviews, and geolocation**, enabling end-to-end analysis of marketplace performance across revenue, retention, delivery operations, and customer experience.

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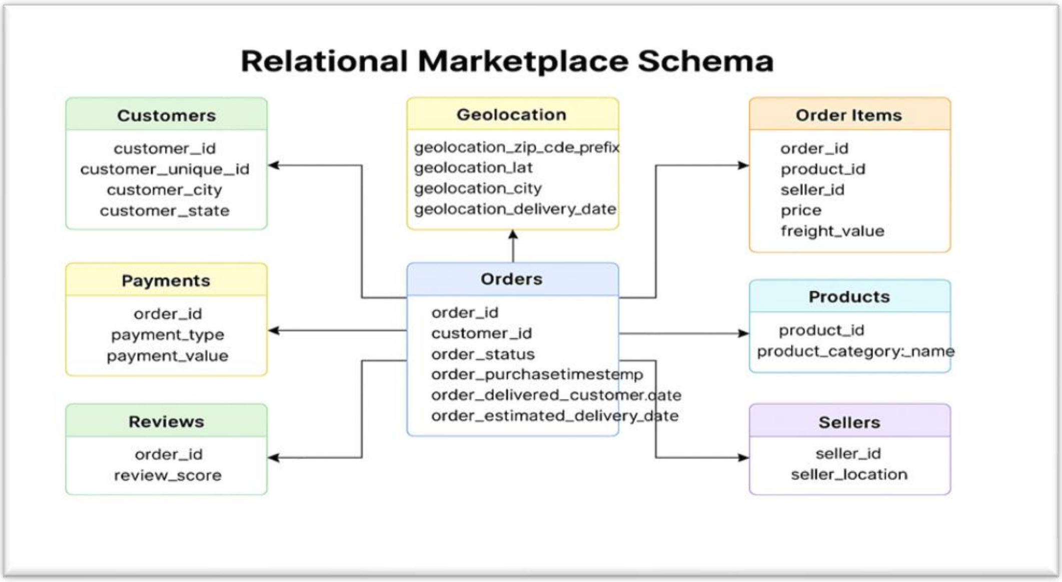
7. Data Understanding

7.1 Schema



1) What the dataset represents (High-level overview)

- The Olist dataset is organized as a **relational marketplace schema**.
- It captures the key entities of an e-commerce operation, including:
  - **orders**
  - **customers**
  - **order items**
  - **payments**
  - **reviews**
  - **products**

- **sellers**
- **geolocation**
- **product category translation**

2) Why the Orders table is the "core" table

- The **orders** table is the **central transactional layer** of the dataset.
- It connects the most important business dimensions:
  - **Commercial** (sales/revenue context)
  - **Customer** (who placed the order)
  - **Operational** (delivery and order lifecycle)
- These connections happen through shared identifiers such as:
  - order_id
  - customer_id

3) Key relationships (How tables link together)

A) Orders → Customers

- **Orders** connects to **Customers** using:
  - customer_id
- This allows you to link each order to the customer who placed it.

B) Orders → Order Items / Payments / Reviews

- **Orders** connects to the following tables using:
  - order_id
- These tables are:
  - **order_items** (what was purchased in the order)
  - **payments** (how the order was paid and payment value)
  - **reviews** (customer feedback / review score for the order)

C) Order Items → Products and Sellers

- **order_items** connects to **products** using:
  - product_id
- **order_items** connects to **sellers** using:
  - seller_id
- This enables analysis of:
  - product performance
  - seller performance
  - basket composition (items bought together)

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4) Supporting / reference tables (Enrichment layers)

A) Geolocation table (Geographic enrichment)

- The **geolocation** table provides a geographic reference layer.
- It maps ZIP code prefixes to location details (lat/lng + city/state).
- This helps enable:
  - map visuals
  - geographic segmentation
  - Location-based analysis

B) Product category translation table (Reporting readability)

- The **product_category_name_translation** table is a **lookup table**.
- It maps Portuguese category names to English labels using:
  - product_category_name
  - product_category_name_english
- Its purpose is to make category reporting **business-friendly and easier to communicate**.

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5) What this schema enables analytically

Because everything is connected through IDs (order_id, customer_id, product_id, seller_id), we can produce an integrated dashboard that combines:

- **Sales performance**
- **Customer behavior**
- **Delivery operations**
- **Customer satisfaction (reviews)**
  in one unified analytical model.

7.2 Grain

The dataset contains multiple levels of granularity, and understanding these levels is essential for designing accurate KPIs and avoiding misinterpretation. The **orders** table is at the **order grain**, where each row represents one order. The **order_items** table is at the **order-item grain**, where each row represents one product line within an order, meaning the same order_id may appear multiple times whenever an order contains more than one item. This distinction is critical because revenue, order counts, retention metrics, and basket analysis operate at different analytical grains and must be modeled accordingly.

The **customers** table is at the **customer entity grain**, while customer-level behavioral analysis in this project is performed using **customer_unique_id**, which provides a more reliable identifier for tracking repeat buyers across multiple orders. The **payments** and **reviews** tables enrich transactions at the **order level**, while the **geolocation** and **product category translation** tables function as **supporting reference layers** rather than transactional fact tables. The geolocation dataset contains repeated ZIP code prefixes tied to latitude and longitude records, while the translation table contains one row per category mapping from Portuguese to English.

7.3 Core Analytical Tables

The project is primarily built on the following **core analytical tables**, which directly support the major business objectives of the dashboard.

1. olist_orders_dataset.csv

This is the central transactional table and the backbone of the analysis. It supports **order trend analysis, delivery performance measurement, order status monitoring, and retention timing** through fields such as order_id, customer_id, order_status, order_purchase_timestamp, order_delivered_customer_date, and order_estimated_delivery_date.

The **orders** table provides the core transactional timeline for the business and is central to **revenue trends, order volume analysis, order status tracking, delivery analysis, and retention timing**. Key columns include:

- order_id
- customer_id
- order_status
- order_purchase_timestamp.
- order_delivered_customer_date.
- order_estimated_delivery_date.

2. olist_customers_dataset.csv

This table supports **customer geography, repeat purchase analysis, customer segmentation, and retention analysis**. The most important field is **customer_unique_id**, which is used to consistently identify the same buyer across multiple orders. Additional relevant fields include customer_id, customer_city, and customer_state.

The most important field for this project is:

- customer_unique_id
  which is used to track repeat buyers across multiple orders. Additional relevant fields include:
- customer_id
- customer_city
- customer_state

3. olist_order_items_dataset.csv

This table is the primary source for **product-level, category-level, seller-level, and basket-level analysis**. It enables the project to examine what was purchased within each order and how transactional value is distributed across products and sellers.

Key columns include:

- order_id
- product_id
- seller_id
- price
- freight_value

4. olist_order_payments_dataset.csv

This table supports **revenue calculation** and **payment behavior analysis**, enabling the project to examine total payment value as well as payment method preferences.

Key columns include:

- order_id
- payment_type
- payment_value

5. olist_order_reviews_dataset.csv

This table is central to the **customer satisfaction** layer of the project. It allows review outcomes to be linked to order and delivery performance, making it possible to analyze the relationship between service quality and customer experience.

Key columns include:

- order_id
- review_score

6. olist_products_dataset.csv

This table enriches item-level transactions with product classification and supports **product and category performance analysis**.

The most relevant columns include:

- product_id
- product_category_name

7. olist_sellers_dataset.csv

This table supports **seller-level and marketplace operations analysis**, particularly where seller location and fulfillment patterns are relevant to performance interpretation.

Key columns include:

- seller_id
- seller location fields

7.4 Supporting / Reference Tables

The following tables are not primary transactional fact tables, but they provide important enrichment for reporting, readability, and geographic interpretation.

8. olist_geolocation_dataset.csv

This table links Brazilian ZIP code prefixes to **latitude, longitude, city, and state** information and serves as a geographic reference layer for **regional enrichment and map-based analysis**. Because it contains multiple records for the same ZIP code prefix, it should be treated as a reference dataset that may require standardization or aggregation before use in the final data model. Its value lies in supporting **geographic segmentation, state-level analysis, and regional visualization**, rather than serving as a direct analytical fact table.

Based on the provided sample, key columns include:

- geolocation_zip_code_prefix
- geolocation_lat
- geolocation_lng
- geolocation_city
- geolocation_state

9. product_category_name_translation.csv

This is a **lookup table** that maps original Portuguese category names to their English equivalents through the fields product_category_name and product_category_name_english. It is not transactional in nature, but it is incredibly important for **dashboard readability, executive reporting, and portfolio presentation quality**, as it allows product and category insights to be presented in a more business-friendly format.

The file shows two columns:

- product_category_name
- product_category_name_english

7.5 Analytical Relevance to Project Objectives

The structure of the dataset is well aligned with the objectives of **Olist 360** because it supports multiple business perspectives within the same project.

- The combination of the following:
  - **orders + payments** support **sales and revenue analysis**;
  - **orders + customers** support **repeat purchase behavior, customer segmentation, customer value concentration, and cohort retention**;
  - **orders + reviews** support **customer satisfaction and delivery impact analysis**
  - **order_items + products + category translation** supports **product performance, category reporting, and basket analysis**.

- The **geolocation** dataset strengthens the geographic dimension of the project by enabling regional enrichment and location-based visualization.

Overall, this data model is not only suitable for descriptive reporting, but also for more advanced business intelligence use cases such as **RFM segmentation, retention analysis, delivery diagnostics, and product association analysis**, which are central to the project's executive and customer intelligence focus.

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8. Methodology

8.1 Methodological Overview

This project followed an end-to-end business intelligence and analytics workflow designed to transform the Olist marketplace data into a structured decision-support solution. The methodology combined **data preparation, analytical modeling, exploratory analysis, feature engineering, dashboard development, and business interpretation** in order to answer strategic questions related to revenue growth, customer retention, delivery performance, product/category performance, and basket behavior.

Rather than relying on raw transactional tables directly, the methodology introduced an intermediate **analytical feature layer** to ensure that all KPIs were calculated at the correct grain. This was critical for avoiding double counting and for producing reliable measures in Power BI, particularly for revenue, customer value, retention, basket metrics, and category-level analysis.

8.2 Workflow Stages

The project was executed through the following analytical stages:

- **Data ingestion and relational understanding**
  The source tables were reviewed as a relational marketplace schema covering orders, customers, products, payments, reviews, sellers, geolocation, and category translation. Analytical relevance, schema structure, and table grain were documented before transformation began.
- **Data quality assessment and cleaning**
  The raw data was validated for key integrity, missing values, duplicates, referential integrity, numeric anomalies, timestamp consistency, and category/geolocation issues. Data quality flags were introduced where appropriate, and cleaned datasets were exported for analytical use.

- **Feature engineering and analytical table creation**
  Business-ready analytical tables were created to support the major use cases of the project. These included order-level, item-level, customer-level, RFM, basket, and cohort feature tables, all built at the correct analytical grain.
- **Exploratory Data Analysis (EDA)**
  The engineered tables were used to analyze revenue and order trends, geography concentration, category performance, delivery quality, satisfaction drivers, customer segmentation, retention patterns, and basket behavior. The purpose of the EDA was to convert cleaned data into business insights and strategic recommendations.
- **Dashboard design and KPI modeling in Power BI**
  The analytical outputs were then translated into a Power BI reporting environment organized by business theme, including Sales & Trends, Geography, Category Performance, Delivery & Satisfaction, Customers / RFM, Cohort Retention, and Basket & Cross-Sell.
- **Business interpretation and recommendation development**
  The final stage translated analytical findings into business recommendations focused on delivery improvement, retention strategy, geographic prioritization, category management, and cross-sell activation.

8.3 Tools and Technologies Used

The project used a combination of tools; each aligned to a specific stage of the analytical workflow:

- **SQL** for relational preparation, integrity checks, and schema validation
- **Python / Jupyter Notebook** for cleaning, feature engineering, EDA, retention preparation, and basket analysis
- **Power BI Desktop** for data modeling, DAX KPI development, dashboarding, and business-facing reporting
- **CSV exports** as the controlled analytical handoff layer between preparation and reporting

8.4 Design Principle: Correct Grain and KPI Integrity

A key methodological principle throughout the project was aligning each KPI with the correct table grain. Revenue and order KPIs were calculated at the **order level**, category metrics at the **item/category level**, customer metrics at the **unique customer level**, cohort metrics at the **cohort-period level**, and basket metrics at the **order basket level**. This principle ensured consistent KPI behavior across filters, comparisons, and dashboard drill-down paths.

8.5 Methodological Outcome

This methodology produced a complete analytics workflow that is both **business-facing** and **technically reproducible**. It enabled the project to move from raw marketplace records to a structured analytical model, validated insights, and an executive-ready Power BI dashboard suite.

9. Data Quality Assessment, Cleaning & Preprocessing.

A. (SQL Phase)

9.1 Data Ingestion & Table Selection

1. **Load the 9 Olist CSV files** into the analytical environment (SQL).
2. **Classify tables by role**:
   - **Core analytical tables**: orders, customers, order_items, payments, reviews, products, sellers.
   - **Reference/enrichment tables**: geolocation and category translation (for geographic mapping and English category reporting).

9.2 Data Cleaning & Processing (before modeling)

To ensure KPIs are trustworthy, the following checks are performed early and repeatedly:

- **Uniqueness & integrity**
  - Uniqueness and referential integrity checks confirmed that the transactional data is structurally sound and suitable for analytical use. The order_id field in the orders table is fully unique, confirming that the dataset is maintained at the correct order-level grain. No duplicate order records were detected.

  In addition, all key relationships across core tables were validated. Every order record is associated with a valid customer, and all related tables (order items, payments, and reviews) reference existing orders. Likewise, all order items reference valid products and sellers. No orphan records or broken relationships were identified, indicating strong referential integrity across the dataset.

- **Missing values**
  - Missing value analysis was performed across all critical identifiers, timestamps, and descriptive attributes. No missing values were detected in primary or foreign key fields such as order IDs, customer IDs, product IDs, or seller IDs, ensuring that relational joins can be safely performed.
  - Two notable cases were identified:
    - A subset of orders contains missing delivery timestamps. These records correspond to orders that were not delivered at the time of data extraction (e.g., canceled or in-progress orders) and therefore represent a valid business condition rather than a data quality issue.
    - **A small number of products are missing category labels**. This represents a genuine data quality gap that **may affect category-level analysis and requires a defined handling rule in subsequent cleaning or modeling steps.**

Issue Description

The presence of missing category values violated the assumptions required for referential integrity enforcement. Since core_products.product_category_name is designed to reference the category master data in ref_category_translation, every product record must contain a valid and standardized category value. Missing or empty values prevented the successful creation of the foreign key constraint and represented a data consistency risk.

Analysis Outcome

A focused examination confirmed that a subset of product records did not have a valid category assignment. These values were not distributed randomly but represented legitimate products that lacked classification during the original data ingestion process. Although these records did not compromise transactional accuracy, they interfered with relational integrity enforcement and dimensional consistency.

Resolution Approach

To resolve the issue without data loss or record exclusion, a controlled remediation strategy was applied. All missing or empty category values were standardized to a single, well-defined placeholder category. This approach ensured that:

- No product records were removed or altered structurally
- Category classification remained explicit and traceable
- Referential integrity rules could be enforced consistently

The reference table was validated and completed accordingly to include the standardized category value, ensuring a complete and authoritative domain for category lookups.

Final Status

Following the remediation:

- All product records contain valid category values
- The category reference domain is complete and consistent
- Foreign key constraints can be enforced without violation.
- The data model maintains both analytical integrity and referential consistency.

This resolution was treated as a data quality correction rather than a modeling change and does not affect downstream analytical logic or business assumptions.

- **Duplicate handling**
    - Comprehensive duplicate checks were conducted to identify potential issues caused by data reloading, faulty joins, or source-system duplication. No duplicate transactional records were detected across unique keys, composite keys, or full-row comparisons.
    - This confirms that the dataset does not contain unintended record duplication and that transaction-level data can be used confidently without risk of double counting or inflation of analytical metrics.

- **Data type standardization**

Before performing any data type conversions, a dedicated validation step was executed to assess whether key timestamp and numeric fields could be safely standardized without parsing or casting failures. This check was designed to detect **invalid datetime strings** and **non-numeric values** that would break downstream transformations, KPI calculations, or analytical models.

The validation results confirmed **zero invalid values** in the tested fields, indicating that the dataset is structurally ready for data type standardization in the subsequent cleaning layer. Specifically, no invalid timestamps were detected in core order and review date fields, and no invalid numeric values were detected in core monetary and installment fields.

Validation outcomes (invalid value counts):

- `order_purchase_timestamp_invalid_timestamp = 0`
- `order_approved_at_invalid_timestamp = 0`
- `review_creation_date_invalid_timestamp = 0`
- `price_invalid_numeric = 0`
- `payment_installments_invalid_numeric = 0`

Implication: Since invalid parsing/casting cases were not observed, the next phase can proceed with converting these columns to proper `DATETIME` and numeric data types with minimal risk of transformation failures.

9.3 Data Modeling (Schema & Grain Control)

Phase 1 — Schema Readiness, Key Strategy & Grain Control

1. Objective of Phase 1

The objective of Phase 1 was to prepare the Olist 360 data model for safe and scalable relationship enforcement. This phase focused on **structural readiness**, **key strategy definition**, and **grain control foundations**, ensuring that the model can support accurate KPIs and future analytical use cases without data duplication or integrity risks.

Phase 1 intentionally focused on **design validation and preparation**, not on enforcing physical foreign key constraints or running analytical calculations.

2. Data Model Structure (Core vs Reference Layers)

The data model is organized into clearly separated layers to ensure clarity of responsibility and analytical correctness:

Core Transactional & Entity Tables

These tables represent operational business data and serve as the analytical backbone:

- `core_orders`
- `core_order_items`
- `core_payments`
- `core_reviews`
- `core_customers`
- `core_products`
- `core_sellers`

Reference (Enrichment) Tables

These tables provide descriptive and contextual enrichment without representing transactions:

- `ref_category_translation`
- `ref_geolocation`

This separation ensures that transactional facts, descriptive entities, and reference data are modeled independently while remaining analytically compatible.

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3. Foreign Key Plan Validation (Schema Compatibility)

Before introducing physical foreign key constraints, a **Foreign Key (FK) plan validation** was performed to verify that all intended relationships are structurally valid.

The validation confirmed that:

1. Join key columns exist in both child and parent tables.
2. Data types and column lengths match exactly across relationships, preventing join instability or FK creation failures.

Across repeated validation runs, **all planned relationships returned Status = OK**, confirming consistent schema compatibility and readiness.

Validated Relationships

- `core_orders.customer_id → core_customers.customer_id` *(varchar(50) ↔ varchar(50))*
- `core_order_items.order_id → core_orders.order_id` *(varchar(50) ↔ varchar(50))*
- `core_payments.order_id → core_orders.order_id` *(varchar(50) ↔ varchar(50))*
- `core_reviews.order_id → core_orders.order_id` *(varchar(50) ↔ varchar(50))*
- `core_order_items.product_id → core_products.product_id` *(varchar(50) ↔ varchar(50))*
- `core_order_items.seller_id → core_sellers.seller_id` *(varchar(50) ↔ varchar(50))*
- `core_products.product_category_name → ref_category_translation.product_category_name` *(varchar(80) ↔ varchar(80))*

This step ensures that the logical model can later be enforced physically without structural conflicts.

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4. Grain Control (Explicit Definition & KPI Alignment)

To prevent metric inflation and analytical inconsistencies, **grain control was explicitly defined during Phase 1**. Grain control identifies what **a single row represents in each table** and determines where each KPI must be calculated.

4.1 Order-Level Grain — `core_orders`

- **Grain:** One row represents one order.
- **Primary identifier:** `order_id`

KPIs calculated at this grain:

- Total Orders
- Delivered Orders
- Canceled Orders
- Late Delivery Rate
- Average Delivery Time
- Orders per Customer

Grain rule:
Order-level KPIs must be calculated exclusively from core_orders. Orders must not be counted from item-level or payment-level tables.

4.2 Order-Item-Level Grain — core_order_items

- **Grain:** One row represents one product line within an order.
- **Characteristic:** One order may have multiple rows.

KPIs calculated at this grain:

- Items Sold
- Product-Level Revenue
- Category-Level Revenue
- Basket Size
- Product & Category Performance

Grain rule:
Item-level tables must not be used to count orders unless aggregated back to distinct `order_id`.

4.3 Payment-Level Grain — core_payments

- **Grain:** One row represents one payment record.
- **Characteristic:** An order may have multiple payments (installments).

KPIs calculated at this grain:

- Revenue by Payment Method
- Installment Analysis
- Payment Method Distribution

Critical grain rule:
Revenue must be aggregated per order_id **before** joining with orders or items to avoid revenue duplication.

4.4 Review-Level Grain — `core_reviews`

- **Grain:** One row represents one customer review.
- **Relationship:** Reviews attach to orders via order_id.

KPIs calculated at this grain:

- Average Review Score
- Reviews per Order
- Satisfaction Rate
- Delivery Performance vs Satisfaction

5. Surrogate Primary Key Strategy (core_reviews)

During PK candidate validation, it was identified that review_id is **not unique** and therefore unsuitable as a natural primary key. To resolve this without altering or discarding business data, a **surrogate primary key** was introduced.

Implementation

- A new auto-increment column review_row_id was added as the **Primary Key** of core_reviews.

Purpose

- Guarantees row-level uniqueness.
- Preserves review_id as a business identifier.
- Ensures order_id remains a **Foreign Key**, not a Primary Key.
- Enables safe indexing and relationship enforcement.

Supporting indexes were created on review_id and order_id to optimize query performance and joins.

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6. Phase 1 Outcome

At the completion of Phase 1:

- Schema compatibility has been validated.
- Grain definitions are explicitly documented.
- Primary key strategies are defined, including surrogate keys where natural uniqueness is not guaranteed.
- The model is protected against duplicated counts and inflated KPIs.

The data model is now structurally and conceptually ready for the next phase, which will focus on:

- Orphan record checks,
- Primary and unique key enforcement across remaining tables,
- Index optimization,
- Optional creation of physical foreign key constraints.

Phase 2 — Orphan Checks & PK Enforcement Readiness

1. Objective of Phase 2

The objective of Phase 2 was to validate **data-level referential integrity** prior to enforcing physical foreign key constraints. While Phase 1 confirmed structural readiness at the schema level, Phase 2 ensures that the **actual data does not violate expected parent–child relationships**.

This phase protects the model from foreign key creation failures, eliminates hidden data quality issues, and guarantees that downstream joins and KPIs are based on consistent relationships.

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2. Orphan Checks Methodology

An **orphan record** is defined as a row in a child table that references a key value that does not exist in the corresponding parent table.

Orphan checks were executed across all critical relationships in the Olist 360 model, covering:

- Orders → Customers
- Order Items → Orders
- Order Items → Products
- Order Items → Sellers
- Payments → Orders
- Reviews → Orders
- Products → Category Translation

To ensure scalability and performance on large transactional tables, orphan detection was optimized using:

- Supporting indexes on foreign key columns
- Binary-safe comparisons to eliminate collation conflicts
- NOT EXISTS logic to reduce execution cost on high-volume tables

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3. Orphan Check Results

During the execution of Phase 2 orphan checks, the validation process initially confirmed that all core parent–child relationships were free from structural orphan records. Specifically, foreign key values across the evaluated relationships correctly referenced existing parent keys, indicating a strong baseline of referential alignment.

However, during the subsequent foreign key enforcement attempts in Phase 3, a limited **data completeness issue** was identified within the product categorization domain. A small number of product records contained **valid category values that were not yet registered in the category**

reference table (ref_category_translation). While these values were not null or invalid, their absence from the reference domain temporarily prevented foreign key enforcement.

This finding did not represent a violation of referential logic but rather highlighted an **incomplete reference domain**, which is a common data quality scenario in analytical datasets.

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4. Actions Performed

To resolve the issue and preserve data integrity without data loss:

- Missing category values were identified through targeted validation queries.
- The reference table (ref_category_translation) was completed by inserting the missing category codes along with their corresponding English translations.
- No transactional or fact data was removed, altered, or reclassified.
- The remediation was treated strictly as a **data completion step**, not a modeling correction.

Following this controlled correction, all orphan checks and referential validations were successfully re-executed with no remaining violations.

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5. Final Phase 2 Outcome

At the conclusion of Phase 2, incorporating the final remediation results:

- ✅ No orphan records remain across all evaluated parent–child relationships
- ✅ Referential integrity is fully confirmed at the data level
- ✅ All foreign key domains are complete and consistent
- ✅ The data model passes all integrity and readiness checks
- ✅ The schema is cleared for physical foreign key enforcement

Phase 2 Status: PASSED

The Olist 360 data model is now fully prepared for Phase 3, which focuses on enforcing foreign key constraints and finalizing referential integrity at the database level.

Phase 3 — Primary and Foreign Key Enforcement

1. Objective of Phase 3

The objective of Phase 3 was to **physically enforce the validated logical data model** by implementing **primary key (PK)** and **foreign key (FK)** constraints across the Olist 360 schema.

This phase represents the final transition from a logically consistent and data-validated schema to a **fully enforced relational model**, where referential integrity is guaranteed directly by the database engine.

2. Primary Key Enforcement Overview

Before enforcing foreign keys, all tables were assessed to ensure that each one had a valid primary key suitable for enforcing relational dependencies.

Design Overview

- Entity and reference tables are identified by **single-column primary keys**.
- Fact tables with multi-row granularity use **composite primary keys** aligned with their business grain.
- **Surrogate primary keys** were introduced in cases where natural identifiers could not guarantee uniqueness at the row level.

This ensured that all tables could safely participate in referential relationships without ambiguity.

3. Relationship & Key Mapping Matrix

The table below summarizes **all enforced relationships**, identifying parent and child roles, primary keys, foreign keys, and notable design considerations.

Phase 3 – Relationship & Key Mapping

#	Parent Table	Parent Primary Key	Child Table	Child Foreign Key	Design Notes
1	core_customers	customer_id	core_orders	customer_id	Orders cannot exist without a valid customer
2	ref_category_translation	product_category_name	core_products	product_category_name	Category standardization via reference domain
3	core_orders	order_id	core_order_items	order_id	Prevents item records without a parent order
4	core_products	product_id	core_order_items	product_id	Ensures items reference valid products
5	core_sellers	seller_id	core_order_items	seller_id	Enforces seller validity per order item
6	core_orders	order_id	core_payments	order_id	Payments always belong to an order
7	core_orders	order_id	core_reviews	order_id	Reviews always linked to a valid order

Additional Key Design Notes

- core_reviews uses a **surrogate primary key (review_row_id)** due to non-unique review identifiers.
- ref_geolocation uses a **surrogate primary key (geolocation_row_id)** because ZIP code prefixes are not unique at the row level.
- These additions ensure row-level identity without altering business semantics.

4. Foreign Key Enforcement Strategy

All foreign key constraints were implemented using a consistent enforcement policy aligned with analytical and governance best practices.

Constraint Rules Applied

- **ON UPDATE CASCADE**
  Ensures identifier changes propagate safely across dependent tables.
- **ON DELETE RESTRICT**
  Prevents deletion of parent records that have dependent child records, protecting historical and analytical integrity.

Each foreign key was applied incrementally and validated at creation time to ensure full compliance with relational constraints.

Geolocation Table Relationship Design Note

The `ref_geolocation` table was not enforced as a physical foreign-key relationship in the SQL relational model.

Reason

The raw `ref_geolocation` table contains multiple records for the same `geolocation_zip_code_prefix`.
This means that `geolocation_zip_code_prefix` is not unique in the raw table and therefore is not suitable to be used directly as a parent primary key in a strict relational schema.

If the raw geolocation table were joined directly with customer or seller tables, one ZIP code could match multiple geolocation records. This may duplicate customer or seller rows during joins and inflate analytical results.

Treatment Decision

Instead of creating a direct foreign-key constraint, the geolocation table will be prepared separately by:

- removing exact full duplicate rows,
- aggregating the table to one row per ZIP code,
- creating a cleaned ZIP-level lookup table named `ref_geolocation_zip`.

Analytical Relationships

After geolocation aggregation, the cleaned ZIP-level table can be joined analytically with:

- `core_customers.customer_zip_code_prefix`
- `core_sellers.seller_zip_code_prefix`

using:

- `ref_geolocation_zip.geolocation_zip_code_prefix`

Final Decision

`ref_geolocation` is treated as an analytical reference table, not as a directly enforced SQL relationship in the raw schema.

Parent Table	Parent Primary Key	Child Table	Child Foreign Key	Design Notes
ref_geolocation_zip	geolocation_zip_code_prefix	core_customers	customer_zip_code_prefix	Analytical ZIP-level lookup after geolocation cleaning and aggregation
ref_geolocation_zip	geolocation_zip_code_prefix	core_sellers	seller_zip_code_prefix	Analytical ZIP-level lookup after geolocation cleaning and aggregation

5. Phase 3 Outcome

At the conclusion of Phase 3:

- ✅ All tables have enforced Primary Keys
- ✅ All planned Foreign Key relationships are physically implemented
- ✅ Parent–child dependencies are enforced at the database level
- ✅ Invalid inserts, updates, and deletions are systematically prevented
- ✅ The schema fully reflects the intended logical data model

Phase 3 Status: PASSED

6. Final Model Enforcement Statement

With Phase 3 completed, the Olist 360 data model is now **fully enforced and governed by the database engine itself**. Referential integrity no longer relies on upstream processes or manual validation, providing a robust, auditable, and production-ready foundation for analytics, reporting, and downstream consumption.

9. Data Quality Assessment, Cleaning, and Preparation

B: In Python

1. Executive Summary

This report documents the data quality assessment, cleaning, and preparation activities performed on the OLIS360 e-commerce dataset. The objective was to validate the dataset's readiness for exploratory data analysis, reporting, dashboarding, and downstream analytical workflows.
The assessment covered file loading, data types, missing values, duplicate records, relationship integrity, numeric validation, date logic validation, categorical consistency, geolocation preparation, and final export of cleaned datasets.
Overall, the dataset was found to be structurally reliable after applying targeted cleaning actions and creating data quality flags where direct modification was not appropriate. The raw data was preserved, while cleaned and analysis-ready versions were exported for future use.

2. Dataset Scope

The assessment covered the following core and reference tables:
Core Tables

- *core_customers*
- *core_orders*
- *core_order_items*
- *core_payments*
- *core_products*
- *core_reviews*
- *core_sellers*

Reference Tables

- *ref_category_translation*
- *ref_geolocation*

A final ZIP-level geolocation reference table, *ref_geolocation_zip*, was also created for safe analytical joins with customer and seller tables. [2]

3. Assessment Methodology

The data quality assessment followed a structured process covering:

1. File loading and parsing validation
2. Data type validation and datetime conversion
3. Missing value analysis
4. Duplicate record analysis
5. Primary key and composite key validation
6. Reference table consistency checks
7. Referential integrity checks
8. Numeric value validation
9. Date business rule validation
10. Categorical value consistency checks
11. Geolocation deduplication and ZIP-level aggregation
12. Export of final cleaned datasets [3]

4. Key Issues Identified and Treatment Decisions

4.1 Date and Timestamp Data Type Issues

Several date columns were initially stored as string/object values and were converted to standardized *datetime64[ns]* format. The affected columns included order timestamps, shipping limit date, and review timestamps. After conversion, all expected date columns were successfully parsed, and no invalid date strings were found.

Treatment Applied

- Converted all expected date columns to *datetime64*
- Preserved missing values as *NaT*.
- Confirmed that missing date values were not caused by parsing failures.

Decision

Date conversion was completed successfully. Missing date values were analyzed separately by business status rather than treated as technical parsing errors.

4.2 Missing Dates by Order Status

Missing order date values were analyzed by order_status. Most missing values were consistent with the order lifecycle. For example, created, approved, processing, invoiced, shipped, unavailable, and canceled orders may naturally have missing delivery-related timestamps.

The only notable exceptions were a small number of delivered orders missing critical timestamps:

Data Quality Flag	**Flagged Records**
dq_delivered_missing_approved_at	14
dq_delivered_missing_carrier_date	2
dq_delivered_missing_customer_date	8
dq_delivered_any_missing_critical_date	23

Treatment Applied

- Original date columns were kept unchanged.
- Data quality flags were created for delivered orders with missing critical timestamps. [9]

Decision

These records should be reviewed or excluded only during delivery-related KPI calculations. No records were deleted during the preparation phase.

4.3 Review Comment Missingness

The review text fields showed high missingness:

Column	**Empty / Blank Percentage**	**Non-Empty Text Percentage**
review_comment_title	88.34%	11.66%
review_comment_message	58.73%	41.27%

This was considered expected behavior because customers can submit ratings without writing textual comments. Reviews without text still contain valid review scores and should not be removed.

Treatment Applied

Created review text availability flags:

Flag	**True Count**	**True Percentage**
has_review_comment_title	11,565	11.66%
has_review_comment_message	40,948	41.27%
has_any_review_text	42,685	43.02%

Insight

Reviews without written text had a higher average score of **4.38**, while reviews with written text had a lower average score of **3.70**. This suggests that customers who write comments may be more likely to provide detailed feedback about issues or dissatisfaction. [14]

Decision

Review records were retained. Text availability flags will be used during EDA to compare customer satisfaction patterns.

4.4 Missing Category Translation

One missing value was found in *ref_category_translation.product_category_name_english*. The missing value belonged to the category *unknown*. [16]

Treatment Applied

- Filled the missing English category translation with *"unknown"*.
- Verified that no missing or blank translation values remained.
- Confirmed that all product categories in *core_products* matched the translation table.

Decision

The category translation issue was fully resolved. No additional category mapping treatment was required.

4.5 Full Duplicate Rows

Full-row duplicate checks were performed across all tables. All tables had zero full duplicate rows except *ref_geolocation*.

Table	Original Rows	Full Duplicate Rows Removed	Duplicate % Removed	Cleaned Rows
ref_geolocation	1,000,163	261,831	26.18%	738,332

Treatment Applied

A cleaned version of the geolocation table was created:

ref_geolocation_clean

Exact full duplicate rows were removed using *drop_duplicates()*, while preserving the original raw table.

Decision

ref_geolocation_clean should be used instead of the raw ref_geolocation table when a deduplicated geolocation reference is required.

4.6 Primary Key and Composite Key Validation

Primary key and composite key checks were performed to distinguish true duplicate issues from naturally repeated values.

Primary Keys Checked

- *core_customers.customer_id*
- *core_orders.order_id*
- *core_products.product_id*
- *core_sellers.seller_id*
- *ref_category_translation.product_category_name*

All checked primary keys were unique. [24]

Composite Keys Checked

- *core_order_items*: *order_id* + *order_item_id*
- *core_payments*: *order_id* + *payment_sequential*

Both composite keys were unique.

Review Table Diagnosis
Although *review_id* and *order_id* had duplicates individually, the combined key *review_id* + *order_id* was fully unique. Therefore, no review records were removed.
Decision
No additional duplicate treatment was required outside the geolocation table.

4.7 Geolocation ZIP-Level Duplication
After full duplicate removal, *ref_geolocation_clean* still contained multiple records per ZIP code. This could cause row multiplication during joins with customer or seller tables.
Diagnosis Results

Metric	Value
Rows in *ref_geolocation_clean*	738,332
Unique ZIP codes	19,015
Extra rows due to ZIP-level duplicates	719,317
ZIP codes appearing more than once	17,823
Maximum rows for a single ZIP code	779

Treatment Applied
A ZIP-level lookup table was created:
ref_geolocation_zip
Aggregation logic:
- Mean latitude
- Mean longitude
- Most frequent city
- Most frequent state

Validation
- Duplicate ZIP codes in *ref_geolocation_zip*: **0**
- Customer rows before join: **99,441**
- Customer rows after join: **99,441**
- Seller rows before join: **3,095**
- Seller rows after join: **3,095**

Decision
ref_geolocation_zip should be used for future joins with customer and seller ZIP codes.

4.8 Referential Integrity Checks
Referential integrity checks confirmed whether child table keys matched valid parent table keys. Most core relationships passed with zero unmatched records.
Passed Relationships
- *orders -> customers*
- *order_items -> orders*
- *order_items -> products*
- *order_items -> sellers*
- *payments -> orders*
- *reviews -> orders*
- *products -> category_translation*

All of these relationships had **0 unmatched child rows**.
Geolocation Relationship Exceptions

Relationship	Unmatched Unique Keys	Unmatched Rows	Unmatched Row %
customers -> geolocation_zip	157	278	0.2796%
sellers -> geolocation_zip	7	7	0.2262%

Treatment Applied

Created geolocation match flags:

Table	Match Count	Match %	Unmatched Count	Unmatched %
Customers	99,163	99.72%	278	0.28%
Sellers	3,088	99.77%	7	0.23%

Decision

No customer or seller records were deleted. The unmatched geolocation records were flagged for later use in geographic EDA.

4.9 Numeric Value Validation

Numeric validation checks were performed to identify values that are technically numeric but not logically valid. [38]

Issues Found

Table	Column	Issue	Invalid Rows	Invalid %
core_payments	*payment_value*	*payment_value <= 0*	9	0.0087%
core_payments	*payment_installments*	*payment_installments <= 0*	2	0.0019%
core_products	*product_weight_g*	*product_weight_g <= 0*	6	0.0182%
core_products	*product_length_cm*	*product_length_cm <= 0*	2	0.0061%
core_products	*product_height_cm*	*product_height_cm <= 0*	2	0.0061%
core_products	*product_width_cm*	*product_width_cm <= 0*	2	0.0061%

Checks With No Issues

- *core_order_items.price <= 0*
- *core_order_items.freight_value < 0*
- *core_products.product_photos_qty < 0*

All returned zero invalid rows.

Treatment Applied

Data quality flags were created for invalid numeric records. Original numeric values were not changed. [41]

Decision

The affected percentages were very small, so records were flagged rather than removed. Further handling can be decided during EDA or modeling.

4.10 Date Business Rule Validation

Date business rules were validated to confirm that event sequences were logically correct.

Issues Found

Table	Rule Violated	Rows With Both Dates	Invalid Rows	Invalid %
core_orders	*order_approved_at > order_delivered_carrier_date*	97,644	680	0.6964%
core_orders	*order_delivered_carrier_date > order_delivered_customer_date*	96,475	20	0.0207%
core_reviews	*review_creation_date > review_answer_timestamp*	99,222	18,689	18.8355%

Treatment Applied

Date logic flags were created:

Table	Flag	Flagged Rows	Flagged %
core_orders	*dq_approval_after_carrier*	680	0.6838%
core_orders	*dq_carrier_after_customer*	20	0.0201%
core_reviews	*dq_review_creation_after_answer*	18,689	18.8355%

Decision

No original timestamps were modified. The flags will be used later during EDA and KPI calculations to exclude or separately analyze affected records.

4.11 Categorical Value Consistency

Categorical value checks were performed on order status, payment type, customer state, seller state, geolocation state, and product category fields.

Key Results

- No missing categorical values were found in reviewed columns.
- No blank or whitespace-only categorical values were found.
- No leading or trailing space issues were detected.
- *order_status* contained only expected values.
- *payment_type* contained only expected values.
- All product categories matched the category translation table.

State Consistency

All customer and seller states existed in the geolocation reference table. The geolocation table contained four states not present in the seller table: AL, AP, RR, and TO. This was not considered a data quality issue, because it only means no sellers were registered in those states.

Decision

No categorical values required modification. Categorical fields are ready for EDA and downstream analysis.

5. Final Clean Dataset Export

The cleaned and analysis-ready datasets were exported successfully to the *clean_exports* folder.

Dataset	Rows	Columns	File Path
core_customers_clean	99,441	6	*clean_exports\core_customers_clean.csv*
core_orders_clean	99,441	17	*clean_exports\core_orders_clean.csv*
core_order_items_clean	112,650	9	*clean_exports\core_order_items_clean.csv*
core_payments_clean	103,886	7	*clean_exports\core_payments_clean.csv*
core_products_clean	32,951	14	*clean_exports\core_products_clean.csv*
core_reviews_clean	99,222	12	*clean_exports\core_reviews_clean.csv*
core_sellers_clean	3,095	5	*clean_exports\core_sellers_clean.csv*
ref_category_translation_clean	74	2	*clean_exports\ref_category_translation_clean.csv*
ref_geolocation_clean	738,332	5	*clean_exports\ref_geolocation_clean.csv*
ref_geolocation_zip	19,015	5	*clean_exports\ref_geolocation_zip.csv*

Export Decision

The raw source files remain unchanged. The exported files represent the cleaned working versions of the datasets and are ready for EDA, SQL loading, Power BI dashboards, reporting, modeling, and further analytical workflows.

6. Final Data Quality Status

After completing the data quality preparation phase:

- Date columns were standardized.
- Missing values were interpreted using business context.
- Critical missing date issues were flagged.
- Review comment missingness was transformed into useful analytical flags.
- Category translation issues were resolved.
- Exact duplicate rows were removed from geolocation data.
- Primary and composite keys were validated.
- Review-level duplicates were diagnosed and confirmed as non-critical at the combined key level.
- Geolocation data was aggregated into a ZIP-level lookup table.
- Referential integrity was validated across the main data model.
- Numeric anomalies were flagged.
- Date sequence anomalies were flagged.
- Categorical fields were validated and confirmed consistent.
- Clean datasets were exported successfully.

Conclusion

The data quality preparation phase was completed successfully. The dataset is now structured, validated, documented, and ready for professional analytical use. No major blocking data quality issues remain for the EDA phase, provided that the created data quality flags are considered during KPI calculation, geographic analysis, and modeling.

10. Exploratory Data Analysis (EDA)

10.1. Executive Summary

This Exploratory Data Analysis (EDA) was conducted to transform the cleaned transactional, customer, delivery, and product data into **business-ready insights** that support strategic decision-making. The analysis was designed to answer core business questions related to **sales growth, customer behavior, geographic concentration, category performance, delivery quality, customer retention, and cross-sell opportunities**. The analytical output also served as the foundation for the Power BI dashboards developed in this project.

Overall, the analysis shows that the business achieved a strong growth phase during 2017 and maintained high revenue levels into 2018. However, this growth appears to be driven primarily by **order volume rather than customer retention**, with very limited repeat purchasing behavior. Revenue is concentrated in a small number of states and categories, while delivery performance emerges as one of the strongest drivers of customer satisfaction. Cross-sell potential exists, but basket behavior remains dominated by single-category purchases.

From a strategic perspective, the key conclusion is that the business is operating as an **acquisition-driven model**, not a retention-driven one. Therefore, the most impactful improvement areas are likely to come from strengthening **customer retention**, reducing **delivery delays**, and increasing **basket breadth** through stronger category recommendations and bundle strategies.

10.2 Analytical Objective and Approach

2.1 Objective

The purpose of the EDA was not merely descriptive profiling. Instead, it aimed to build a **decision-support layer** that explains how revenue is generated, where operational weaknesses exist, which customers create the most value, and how these insights can be monitored through dashboards. The analysis focused on the following themes: revenue and order trends, geography, product/category performance, delivery and customer satisfaction, customer segmentation, cohort retention, and basket behavior.

2.2 Analytical Methodology

Before running the EDA, a dedicated **feature engineering layer** was created to ensure that every metric was calculated at the correct analytical grain. Order-level metrics were generated at the order_id level, item/category metrics at the order-item level, customer metrics at the customer_unique_id level, and basket metrics at the order level. This design was critical to avoid double counting and to ensure that Power BI KPIs remained accurate. All engineered feature tables passed grain validation checks.

The analysis then proceeded in a structured way across multiple business dimensions. Revenue and orders were explored over time and geography, product categories were evaluated using revenue, order count, reviews, and late-delivery rates, and customer behavior was examined using RFM segmentation, cohort retention, repeat behavior, revenue concentration, and basket patterns. This multi-layer approach allowed the project to connect commercial performance with customer and operational drivers.

10.3 Revenue and Order Trends

The business experienced a clear growth phase during **2017**, followed by sustained strong performance throughout much of **2018**. The highest-performing month was **November 2017**, which generated **7,544 orders**, **1,194,882.80** in revenue, and an **average order value (AOV) of 158.39**. This month ranked first in both revenue and order volume, making it the strongest single month in the observed period.

Between **January 2018 and August 2018**, revenue remained consistently high, with several months approaching or exceeding **1 million** in revenue. At the same time, AOV stayed relatively stable, generally within the **150–170** range. This indicates that the main driver of revenue growth was an increase in the **number of orders**, rather than customers spending significantly more per transaction. In other words, growth was primarily volume-driven.

A few months showed abnormally low activity — particularly **2016-09, 2016-12, 2018-09, and 2018-10** — suggesting partial or low data coverage. These periods should not be used to infer seasonality or normal business performance. For reliable trend analysis, the recommended window is **2017-01 to 2018-08**.

Board implication: the business has demonstrated strong acquisition and order volume growth, but not necessarily stronger customer monetization per order. Growth quality therefore depends on whether the company can retain and reactivate those customers over time.

10.4 Geographic Performance

Revenue is heavily concentrated in a small number of states. The dominant market is **SP**, which generated **5,998,226.96** in revenue from **41,746 orders**, representing **37.47% of total revenue** and **41.98% of all orders**. The next two leading states, **RJ** and **MG**, bring the combined contribution of the top three states to **62.56% of total revenue**. The top ten states together account for **87.39%** of revenue, confirming that sales are strongly concentrated in a few core markets.

At the same time, smaller states such as **PB, RO, AL, and PA** show **higher AOVs**, each exceeding **200**, despite lower order volumes. This suggests a second type of opportunity: while large states are volume-driven, smaller markets may offer higher-value transaction potential if demand can be scaled.

Board implication: the business should treat geography as a strategic segmentation variable. Large states need volume optimization and operational protection, while smaller high-AOV markets deserve focused investigation to understand product mix, freight behavior, and premium demand potential.

10.5 Product and Category Performance

Category analysis shows that revenue is driven by a limited group of top-performing categories. The top categories by revenue are **Health & Beauty (1.44M)**, **Watches & Gifts (1.31M)**, **Bed Bath & Table (1.24M)**, **Sports & Leisure (1.16M)**, and **Computers & Accessories (1.06M)**. Altogether, the **top 10 categories account for 62.34% of total revenue**, while the **top 3 account for 25.18%**, indicating moderate concentration at the category level.

The analysis also highlights two distinct commercial patterns. Some categories, such as **Bed Bath & Table**, generate revenue largely through **high order volume**, while others, such as **Watches & Gifts**, depend more on **higher AOV**. This distinction is important for assortment planning and promotional design, as volume-driven categories require different tactics from value-driven ones.

Not all categories perform equally well in customer experience. Some categories exhibit both lower review scores and higher late-delivery rates. Examples include **Office Furniture** (average review around **3.62**), **Audio**, and **Home Comfort**, while categories such as **Audio** also show particularly high late delivery rates (around **11.7%**). This suggests that product performance should not be evaluated by revenue alone; operational reliability and customer experience must be incorporated into the decision framework.

Board implication: category strategy should distinguish between core revenue categories, high-value categories, and problematic categories that require operational or product-quality investigation.

10.6 Delivery Performance and Customer Satisfaction

Delivery quality emerged as one of the strongest business drivers in the analysis. The overall late delivery rate is **6.57%**, while the overall average review score is **4.09**. On the surface, a 6.57% late rate may appear manageable. However, its effect on customer satisfaction is severe. Orders delivered on time average a review score of **4.21**, while late deliveries dropped sharply to **2.27** — a decline of nearly **two full rating points**.

The relationship becomes even more serious when delays are grouped by severity. Reviews remain acceptable for very small delays, but once delays exceed **3 days**, satisfaction deteriorates sharply. Average review scores fall to approximately **2.11** for **4–7 days late**, **1.67** for **8–14 days late** and remain very low thereafter. This strongly supports the operational threshold that delays longer than **3 days** should be treated as business-critical.

Delivery issues are not evenly distributed. High-delay states include **AL (20.58%)**, **MA (16.73%)**, and **SE (14.57%)**, while certain categories — including **Audio**, **Home Comfort**, and **Office Furniture** — also exhibit elevated delay rates. This indicates that customer dissatisfaction is frequently driven by logistics performance rather than product quality alone.

Board implication: reducing delivery delays, especially delays beyond three days, is likely one of the highest impact levers for improving customer satisfaction and protecting brand perception at scale.

10.7 Customer Segmentation and Loyalty

RFM segmentation shows that the customer base is dominated by low-frequency, low-retention behavior. The largest segments are **At Risk (23.95%)**, **Loyal Customers (17.11%)**, **Hibernating (16.05%)**, and **Potential Loyalists (16.03%)**, while **Champions** represent only **6.94%** of customers. Although Champions generate the highest value per customer (**317.71**), they contribute only **13.23% of total revenue** because their population is relatively small.

Revenue is distributed in a surprising way: a large share comes from customers who are not strongly retained. In fact, **At Risk** customers contribute **24.24%** of total revenue, more than any other segment. This indicates that a significant share of current revenue is coming from customers who may not return.

Repeat purchase behavior is extremely limited. Only **3.12%** of customers placed more than one order, while **96.88%** are one-time customers. Repeat customers spend almost **twice as much** on average (**314.99**) compared to one-time customers (**161.82**), but they are too few to materially shift the overall business model.

Board implication: the core business challenge is not customer acquisition but **customer retention**. Long-term value growth will likely depend more on repeat-purchase strategy than on continuing to acquire new one-time buyers.

10.8 Cohort Retention and Basket Behavior

Cohort analysis confirms the retention problem at a structural level. Across nearly all cohorts, retention falls **below 1% immediately after Month 0**, and remains extremely low in subsequent months. There is no visible improvement in later cohorts, which suggests that churn is not a temporary issue but a persistent business pattern.

Basket analysis shows that most baskets are still **single-category baskets**, while only a small number contain multiple categories. However, multi-category baskets produce **higher average order values**, indicating that cross-sell expansion could increase revenue efficiency. Several strong cross-sell rules exist, such as high-confidence and high-lift relationships in niche category pairs, providing the basis for recommendation and bundling strategies.

Board implication: improving retention and broadening baskets are two underexploited growth levers. While customer loyalty is weak, the data suggests cross-sell opportunities with measurable revenue upside.

10.9 Strategic Recommendations

Based on the EDA, the following strategic priorities are recommended:

- **Improve delivery reliability**, especially beyond the **3-day delay threshold**, as this has a direct and severe impact on customer satisfaction.
- **Shift from acquisition-only growth to retention-led growth**, with initiatives aimed at converting one-time buyers into repeat customers.
- **Protect high-volume markets** such as SP, RJ, and MG, while investigating high-AOV states as potential premium-growth opportunities.
- **Segment category strategy** into: core revenue generators, high-value premium categories, and operationally risky categories that need service improvement.
- **Develop cross-sell and bundle logic** from basket rules to increase multi-category orders and raise AOV.

10.10 Use of EDA in Power BI Dashboards

The EDA outputs directly informed the design of the Power BI solution. The dashboards were structured around the main analytical themes: **Sales & Trends**, **Geography**, **Category**, **Delivery & Satisfaction**, **Customers (RFM)**, **Cohort Retention**, and **Basket & Cross-sell**. Each page was built on the correctly engineered analytical grain, ensuring reliable KPIs, filters, comparisons, and drilldowns for business monitoring.

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11. Analytical Validation Summary

To strengthen analytical credibility, selected findings were validated using non-parametric statistical testing and effect-size interpretation where appropriate. This step was included to confirm that the most important observed patterns were not only visually apparent in the dashboards, but also analytically meaningful.

Validated analytical conclusions

- **Late delivery has a materially negative impact on customer satisfaction.**
  Orders delivered late showed significantly lower review scores than on-time orders, with a large observed effect size.
- **Delivery delays beyond 3 days represent a critical satisfaction threshold.**
  The analysis showed that review scores deteriorate sharply once delay duration exceeds three days, with both statistical significance and strong practical effect.

- **Repeat customers are significantly more valuable than one-time customers.**
  Although repeat customers represent a very small share of the customer base, they generated substantially higher customer value on average, with a large practical difference.
- **Basket expansion is commercially meaningful.**
  multi-category baskets showed substantially higher average order value than single-category baskets, supporting the existence of practical cross-sell potential.
- **Customer segments differ meaningfully in value and behavioral quality.**
  RFM-based customer groups were not only descriptively distinct, but also analytically separable in terms of customer value contribution.

Interpretation

These validation checks reinforce confidence in the project's core conclusions. The evidence supports the strategic focus on **delivery reliability**, **retention improvement**, **segment-based targeting**, and **basket expansion** as the highest-impact levers for business improvement.

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12. Analytical Feature Layer

12.1 Purpose of the Feature Layer

Following data cleaning and preprocessing, a dedicated **analytical feature layer** was created to transform the raw transactional data into **business-ready metrics** suitable for executive reporting, exploratory analysis, and dashboard design. This layer ensured that each KPI was calculated at the correct level of detail and that the analysis could be performed without duplication or distortion.

The feature layer was designed to support the full analytical scope of the project, including **sales trends, geography performance, category performance, delivery quality, customer segmentation, cohort retention, and basket analysis**. In practice, it served as the bridge between raw relational data and decision-oriented reporting.

12.2 Feature Tables Created

The following feature tables were engineered and validated during the project:

- **orders_features**: one row per order_id, used for revenue, orders, time analysis, and delivery KPIs.
- **order_items_category_features**: one row per order item, used for category-level and product-level analysis.
- **customer_features**: one row per customer_unique_id, used for customer value, repeat behavior, and segmentation analysis.
- **rfm_base / rfm_scored**: customer-level tables used for recency, frequency, monetary scoring and RFM segmentation.
- **basket_features**: one row per order with item records, used for basket size, unique categories, and cross-sell analysis.
- **cohort_retention**: cohort-period summary table used for retention analysis and retention heatmaps.

12.3 Analytical Design Principle

A central design principle in the feature layer was the enforcement of the **correct analytical grain**. Order-level metrics were calculated only at the order level, item-level metrics at the order-item level, and customer-level metrics at the unique customer level. This prevented double counting and ensured that Power BI measures, filters, and drill-downs remained reliable. All feature tables passed grain validation checks.

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13. Analytical Model and Dashboard Design

13.1 Modeling Approach

The reporting model was built to support an executive Power BI solution organized around the main business themes identified in the project. The model combined **fact-like analytical tables** (such as order-level and item-level features) with supporting dimensional tables for customers, categories, geography, and time. This structure enabled consistent KPI calculation, easier drill-down, and controlled filtering behavior across pages.

The model was also intentionally designed to avoid ambiguity in relationships. For example, geolocation was handled through ZIP-level reference tables, and summary-level analytical tables such as cohort retention and cross-sell rules were kept logically separate whenever their grain differed from the main transactional model.

13.2 Dashboard Design Philosophy

The dashboard suite was designed as a **decision-support environment**, not merely a collection of charts. Each page was built around a specific business question, with a consistent navigation system, executive KPI cards, focused visual layouts, and slicers aligned to the analytical purpose of the page. The dashboard screenshots show separate pages for **Sales & Trends, Geography, Category, Delivery, Customers, Cohort Retention, and Basket & Cross-Sell**, each following a consistent visual style and navigation logic.

13.3 Dashboard Themes

The analytical model supported the following dashboard themes:

- **Executive Overview**: high-level KPIs and summary business performance.
- **Sales & Trends**: revenue, orders, growth patterns, and moving averages.
- **Geography**: revenue, orders, and operational performance by state.
- **Category Performance**: revenue concentration, AOV, delivery issues, and customer satisfaction by category.
- **Delivery & Satisfaction**: late delivery behavior and its impact on review scores.
- **Customers / RFM**: segment size, value, and revenue contribution by customer segment.
- **Cohort Retention**: retention heatmaps and cohort trends over time.
- **Basket & Cross-Sell**: recommendation rules, rule strength, and product/category association insights.

This design ensured that board members and business users could move from **high-level summaries** to **functional diagnostics** without losing context.

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14. Key Business Insights

14.1 Revenue Growth Was Strong but Volume-Driven

The business experienced a strong growth phase during **2017**, reaching its peak in **November 2017**, with **7,544 orders**, **1,194,882.80** in revenue, and **158.39** AOV. Through most of **2018**, revenue remained consistently high, but AOV stayed relatively stable in the **150–170** range. This indicates that revenue growth was driven mainly by **higher order volume**, not by a significant increase in spending per order.

14.2 Revenue Is Highly Concentrated in a Few States

The business is geographically concentrated. The state of **SP** alone accounts for **37.47%** of total revenue and **41.98%** of all orders. Together, **SP, RJ, and MG** generate **62.56%** of total revenue, and the top ten states generate **87.39%**. At the same time, several smaller states exhibit much higher AOVs, which suggests untapped premium-value opportunities outside the largest volume markets.

14.3 Category Performance Is Uneven

Revenue is supported by a limited number of strong categories. The top revenue drivers include **Health & Beauty (1.44M)**, **Watches & Gifts (1.31M)**, and **Bed Bath & Table (1.24M)**. The **top 10 categories generate 62.34% of total revenue**, confirming meaningful category concentration. The analysis also showed that some categories are volume-driven, while others are value-driven, which has important implications for pricing, promotion, and assortment strategy.

14.4 Delivery Quality Is a Major Satisfaction Driver

Although the overall late delivery rate is only **6.57%**, late delivery has a disproportionate impact on customer satisfaction. Orders delivered on time have an average review score of **4.21**, while late deliveries fall to **2.27**. Delays of more than **3 days** are associated with a particularly sharp decline in reviews, indicating that delivery timeliness is one of the strongest operational drivers of customer experience.

14.5 The Business Is Highly Acquisition-Driven

Customer analysis shows that the business currently depends heavily on **one-time purchase behavior**. Only **3.12%** of customers placed more than one order, while **96.88%** are one-time buyers. Repeat customers generate almost **double the average value per customer**, but they are too few to materially shift the current business model. This suggests that long-term value creation is constrained more by weak retention than by insufficient acquisition.

14.6 Revenue Is Not Led Mainly by Loyal Customers

RFM analysis revealed a surprising pattern: the largest contributor to revenue is not the “Champions” segment, but the **At Risk** segment, which contributes **24.24%** of total revenue. Although **Champions** generate the highest value per customer (**317.71**), they represent only **6.94%** of customers. This means the business is generating significant revenue from customers who are not being retained effectively.

14.7 Retention Is Structurally Weak

Cohort analysis showed that retention drops sharply after the first purchase. Retention falls **below 1% by Month 1** for most cohorts and remains extremely low afterward. Later cohorts do not show meaningful improvement, indicating that the retention issue is not temporary or seasonal, but structural. This is one of the most important findings of the project.

14.8 Basket Expansion Represents a Growth Opportunity

Most transactions are still single-category baskets, but multi-category baskets generate significantly higher average order values. Basket analysis and rule mining also identified meaningful association patterns between selected categories, which opens the door for stronger recommendation logic, bundling, and “frequently bought together” strategies.

15. Strategic Recommendations

15.1 Reduce Delivery Delays as a Top Operational Priority

Because customer satisfaction drops sharply when deliveries are late—especially beyond **3 days**—delivery reliability should be treated as a high-priority operational KPI. The business should investigate logistics bottlenecks in high-delay states and high-risk categories and integrate delivery timeliness into routine performance monitoring.

15.2 Shift Commercial Focus from Acquisition-Only to Retention-Led Growth

The current growth model depends heavily on attracting new one-time buyers. The analysis strongly suggests that revenue sustainability will improve more through **retention improvement** than through continued acquisition alone. The company should therefore prioritize reactivation and second-purchase strategies for **At Risk** and **Potential Loyalist** customers.

15.3 Protect Core Markets While Investigating Premium-Value Markets

Large states such as **SP, RJ, and MG** should be protected and optimized as core revenue markets. At the same time, smaller high-AOV states should be analyzed further to understand whether their purchasing behavior can be scaled through targeted category, pricing, or service strategies.

15.4 Differentiate Category Strategy

Categories should not be managed uniformly. Core revenue categories should be protected, low-rated categories should be investigated, and high-delay categories should be reviewed from an operational and fulfillment perspective. Category strategy should explicitly distinguish between **high-volume**, **high-value**, and **operationally risky** groups.

15.5 Activate Basket and Cross-Sell Opportunities

Although most baskets are narrow, the presence of strong category association rules suggests meaningful cross-sell potential. Recommendation engines, bundles, and product pairing strategies should be piloted to increase basket breadth and average order value.

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16. Risks, Assumptions, and Limitations

16.1 Data Coverage Constraints

Some months in the timeline showed very low activity and appear to reflect partial or low-coverage periods. These months were treated cautiously in trend analysis, and stable conclusions were primarily drawn from the period **2017-01 to 2018-08**.

16.2 Basket Rules Depend on Historical Co-occurrence

Cross-sell rules are based on historical basket associations. As a result, they reveal statistical co-occurrence patterns rather than causal relationships. Rules with low support may still show high lift, but may not be scalable in practice.

16.3 Cohort Retention Is Extremely Low

The cohort analysis reveals very weak retention. While this is analytically important, it also means that later-month retention metrics may be highly sensitive to small changes in cohort activity. The interpretation should therefore focus more on the structural pattern than on any individual late-month fluctuation.

16.4 One-Time Behavior Dominates the Customer Base

Because repeat purchase behavior is extremely limited, customer-level value metrics should be interpreted in the context of a business that is still largely acquisition-driven. This affects the practical weight of loyalty metrics in the current business model.

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17. Expected Business Impact

If the findings of this project are operationalized, the business can improve performance in several meaningful ways. First, reducing delivery delays—especially delays beyond three days—should directly improve customer satisfaction and potentially reduce negative reviews. Second, stronger retention management can increase customer lifetime value by converting one-time buyers into repeat buyers. Third, better segmentation and category prioritization can improve the allocation of commercial attention and operational resources. Finally, basket expansion strategies can raise average order value and make revenue growth more efficient.

In addition to the analytical insights themselves, the project delivers a complete Power BI dashboard environment that enables management to monitor performance across revenue, logistics, customer behavior, category health, retention, and cross-sell opportunities in a single reporting framework.

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18. Implementation Roadmap / Next Steps

18.1 Immediate Actions (0–30 Days)

- Deploy the Power BI dashboards for executive and operational monitoring.
- Use delivery metrics as an immediate alerting mechanism, especially for states and categories with high late rates.
- Validate business ownership for each dashboard theme (Sales, Operations, Customer, Category).

18.2 Short-Term Actions (30–90 Days)

- Launch targeted retention interventions for **At Risk** and **Potential Loyalist** segments.
- Investigate the operational drivers of late delivery in both high-delay states and high-risk categories.
- Pilot selected cross-sell rules for recommendation or bundling use cases.

18.3 Medium-Term Actions (90+ Days)

- Track the evolution of RFM segment sizes over time to monitor whether retention improves.
- Introduce retention as a core management KPI, supported by cohort monitoring and repeat purchase tracking.
- Expand the analytical model to include experimentation outcomes (promotion tests, fulfillment interventions, or CRM campaigns).

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19. Conclusion

This project successfully transformed raw e-commerce data into a **decision-oriented analytical framework** supported by a structured feature layer, validated EDA, and a fully organized Power BI dashboard suite. The analysis demonstrates that the business is growing, but that this growth is currently driven more by acquisition and order volume than by retention, loyalty, or basket expansion. Revenue is concentrated geographically and categorically, while delivery performance plays a decisive role in customer satisfaction.

The most important strategic conclusion is that the next stage of business maturity should focus on **retention improvement, logistics reliability, and basket development**. If these are addressed successfully, the company is likely to improve not only top-line growth, but also revenue quality, customer lifetime value, and long-term commercial sustainability.

Appendix A – Dashboard Screenshots / Visual Outputs

A.1 Purpose of the Dashboard Appendix

This appendix provides the visual evidence of the analytical solution delivered through Power BI. The dashboard suite was designed to translate the project's analytical outputs into an executive reporting environment that supports decision-making across commercial performance, customer behavior, operations, retention, and cross-sell opportunities. Each dashboard page focuses on a distinct business theme while maintaining a consistent navigation structure, visual identity, and KPI logic.

The dashboards were built on top of the engineered analytical layer created during the project, ensuring that all measures are calculated at the appropriate grain and can be used reliably in drill-down, filtering, and time-based analysis. The visuals included in this appendix are therefore not isolated report pages, but part of a linked analytical system designed for business monitoring.

A.2 Dashboard Pages Included

Figure A1 – Executive Overview

The Executive Overview dashboard serves as the landing page of the reporting environment. It was designed to provide a concise, board-level snapshot of the business using high-level KPIs and summary visuals. Its purpose is to give decision-makers an immediate understanding of overall performance before moving into more specialized analytical pages.

Key elements displayed

- Headline KPIs summarizing business performance
- High-level performance comparison indicators
- Quick access to deeper analytical pages through dashboard navigation
- Summary visuals that connect sales, customer, and operational perspectives

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Figure A2 – Sales & Trends Dashboard

The Sales & Trends dashboard presents the time-based commercial story of the business. It was designed to visualize revenue and order volume evolution across months, quarters, and years, and to distinguish between volume-driven growth and spend-per-order stability. This page supports executive review of trend performance, seasonality, and business momentum.

Key outputs represented

- Revenue trend over time
- Order volume trend over time
- AOV tracking and trend smoothing
- Time-based growth interpretation and directional comparison

Figure A3 – Geography Dashboard

The Geography dashboard translates revenue, order volume, and customer performance into a spatial context. It focuses on identifying which states drive the highest commercial contribution and where order value or operational risk differs across regions. This page is particularly important because the analysis showed strong revenue concentration in a small number of states.

Key outputs represented

- Revenue by state
- Orders by state
- AOV by geography
- Geographic concentration of sales and performance disparities across regions

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Figure A4 – Category Performance Dashboard

The Category dashboard was designed to show which product categories drive revenue, volume, customer value, and potential underperformance. It supports both commercial and operational decisions by combining category revenue concentration with satisfaction and operational perspectives.

Key outputs represented

- Top revenue-driving categories
- Relative category performance comparison
- Category value concentration
- Revenue contribution patterns and category-level prioritization logic

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Figure A5 – Delivery & Satisfaction Dashboard

The Delivery dashboard focuses on the relationship between logistics execution and customer experience. It was built because the EDA demonstrated that delivery timeliness is one of the strongest drivers of review-score deterioration. This page is designed to support operational action by identifying where delays occur and how strongly they affect satisfaction.

Key outputs represented

- Late delivery rate monitoring
- Delay bucket analysis
- Delivery impact on review scores
- Geographic or categorical delivery performance risk tracking

Figure A6 – Customers / RFM Dashboard

The Customers dashboard converts customer-level behavior into business-friendly segments. It combines customer counts, revenue contribution, and RFM-based interpretability in order to distinguish valuable, loyal, at-risk, and dormant customer groups. This visual layer supports both executive understanding and commercial targeting decisions.

Key outputs represented

- Customer count and customer value metrics
- Revenue distribution by segment
- State-level segment distribution
- RFM segment contribution and customer portfolio structure

Figure A7 – Cohort Retention Dashboard

The Cohort Retention dashboard presents customer retention as a time-sequenced behavioral pattern. It was designed to separate acquisition from retention by showing how customer cohorts perform after their first purchase month. This dashboard is essential because the analytical results showed structurally weak retention across almost all cohorts.

Key outputs represented

- Cohort retention KPIs
- Retention curves by cohort
- Retention matrix / heatmap outputs
- Month-based retention comparisons and churn visibility

Figure A8 – Basket & Cross-Sell Dashboard

The Basket & Cross-Sell dashboard was developed to visualize association-rule outputs and category recommendation opportunities. It supports commercial action by identifying category pairs that may improve average basket value through recommendation logic or bundling strategies. This page complements the customer and category dashboards by focusing specifically on **basket expansion**.

Key outputs represented

- Cross-sell rule recommendation view
- Rule strength metrics such as confidence, lift, support, and pair frequency
- Score-based recommendation ranking
- Business interpretation aid for cross-sell opportunities

A.3 Summary of Dashboard Value

Taken together, the dashboard suite provides a unified reporting environment that links **sales performance, geography, delivery quality, customer intelligence, retention, and basket behavior** in one decision-support framework. This appendix documents the visual outputs that operationalize the project's analytical findings and make them accessible to management.

Appendix B – Technical Deliverables

B.1 Purpose of the Technical Deliverables Appendix

This appendix lists the main technical assets delivered as part of the project. These materials collectively support reproducibility, auditability, future dashboard maintenance, and knowledge transfer. The technical package was designed not only to support the current reporting solution, but also to serve as a reusable analytical foundation for future extensions.

B.2 Files Included in the Delivery Package

1. Power BI Report File (.pbix)

The Power BI report file contains the complete interactive dashboard solution, including the data model, measures, filters, navigation logic, and all report pages. This file is the final reporting artifact intended for executive and business-user consumption. It includes the dashboard pages documented in Appendix A and reflects the KPI logic built from the analytical feature layer.

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2. Python / Jupyter Notebook

The notebook contains the analytical workflow used to prepare and validate the project outputs, including:

- Data loading and preparation
- Feature engineering
- Grain validation
- Revenue and trend analysis
- Geographic analysis
- Category performance analysis
- Delivery and customer satisfaction analysis
- RFM segmentation
- Cohort retention analysis
- Basket and cross-sell preparation

This notebook serves as the **technical backbone** of the project and provides transparency around the transformation and analytical logic applied before dashboard development.

3. SQL Scripts

The SQL scripts document the upstream extraction and preparation logic used prior to the Python and Power BI stages. These scripts are important for reproducibility and demonstrate how the project moved from a relational source structure into a business-ready analytics environment.

Typical SQL deliverables include

- Data extraction queries
- Table joins and relational preparation
- Cleaning or pre-aggregation logic
- Export preparation for analytical processing

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4. Cleaned CSV Files

The cleaned CSV files represent the curated exports used in the analytical process and reporting model. These files provide a structured, stable input layer for Python analysis and Power BI modeling.

Examples of delivered analytical datasets include:

- core_customers_clean.csv
- core_orders_clean.csv
- core_order_items_clean.csv
- core_payments_clean.csv
- core_products_clean.csv
- core_reviews_clean.csv
- core_sellers_clean.csv
- ref_category_translation_clean.csv
- ref_geolocation_zip.csv

These cleaned files were essential for preserving referential consistency and for supporting downstream feature engineering.

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5. Analytical Feature Exports

In addition to the cleaned raw-layer tables, the project generated several feature-level and summary tables used for reporting and analysis. These analytical assets form the main reusable output of the project.

Key feature-level deliverables include:

- orders_features
- order_items_category_features
- customer_features
- rfm_base / rfm_scored
- basket_features
- cohort_retention

These tables support reliable KPI construction and reduce future development effort, because they encode business logic directly into reusable analytical structures.

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6. Image Files / Exported Visual Outputs

The project also includes exported image files and dashboard screenshots used for documentation, presentations, and the final written report. These image outputs provide a static, presentation-ready view of the dashboard pages and key visuals created during the project.

B.3 Deliverable Readiness and Reusability

The overall technical package is designed to be **reproducible**, **auditable**, and **extendable**. The cleaned inputs, engineered outputs, analytical notebook, and dashboard file together provide a complete business intelligence workflow from raw data to executive reporting. This structure allows future analysts or business users to reproduce KPIs, extend dashboards, refine feature logic, or add new analytical modules without rebuilding the project from scratch.

B.4 Technical Deliverables Summary

The technical delivery package can be summarized as follows:

- **Business-facing artifact:** Power BI dashboard suite (.pbix)
- **Analytical workflow artifact:** Jupyter notebook with transformation and EDA logic
- **Data preparation artifact:** SQL scripts and cleaned CSV exports
- **Reporting support artifact:** dashboard screenshots and final-report images

Together, these deliverables provide a complete analytical project package suitable for business review, technical validation, and future maintenance.